

ENGINEERING PHYSICS

(BT-201)

UNIT 2

Wave Optics & Interference

■ TOPICS COVERED

1. *Newton's Ring*
2. *Rayleigh Criterion (Resolving Power)*
3. *YDSE & Fringe Width*
4. *Mach-Zehnder Interferometer*
5. *Michelson Interferometer*



fig — interference rings (for reference)

Newton's Ring

Newton's Rings is a classic example of interference in a thin film of variable (non-uniform) thickness. When a plano-convex lens is placed on a flat glass plate and lit with monochromatic light from above, a series of bright and dark concentric circular rings are seen around the point of contact.

► Experimental Setup

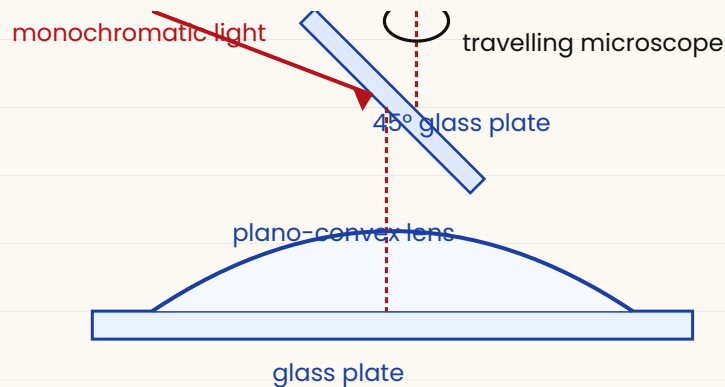


fig — Newton's Ring experimental setup (side view)

A thin air film is formed between the curved lens surface and the flat plate. Its thickness (t) is zero at the centre and increases gradually as we move outward. Interference occurs between light reflected from the top of the air film (lens surface) and the bottom of the air film (glass plate surface).

► Why Rings, and Why Circular?

Points of equal air-film thickness lie on a circle (because the lens surface is spherical) — so all points of equal path-difference form a circle. This is why we see concentric circular fringes instead of straight fringes.

★ **EXAM TIP: "Why are Newton's rings circular?" is a common 2-mark viva/theory question — answer with the "equal thickness locus is a circle" idea above.**

► Condition for Bright and Dark Rings

Path difference for the reflected system = $2\mu t \cos\theta + \lambda/2$ (the extra $\lambda/2$ comes from an extra phase change of π when light reflects from a denser medium at the glass plate — this is why the centre appears DARK, not bright, unlike a normal thin film).

For near-normal incidence ($\theta \approx 0$) and air film ($\mu=1$): Path difference = $2t + \lambda/2$

Condition	Result
$2t = n\lambda \quad (n = 0, 1, 2, \dots)$	Dark ring (destructive interference)
$2t = (2n-1)\lambda/2$	Bright ring (constructive interference)

► Diameter of Rings (Using Geometry, R = radius of lens curvature)

From simple geometry of a circle: $t = r^2/2R$, where r is the distance of the ring from the centre. Combining with the conditions above:

$$D_n (\text{dark}) = \sqrt{4n\lambda R} \quad | \quad D_n (\text{bright}) = \sqrt{2(2n-1)\lambda R}$$

★ **EXAM TIP:** Note that $D_n \propto \sqrt{n}$ — rings get closer together as n increases (they are NOT equally spaced!). This is a favourite numerical/short-answer point.

► Application: Finding Wavelength λ of Light

Measure diameters of the p^{th} and q^{th} dark rings (D_p, D_q) using the travelling microscope. Then:

$$\lambda = (D_q^2 - D_p^2) / 4R(q-p)$$

► Application: Finding Refractive Index μ of a Liquid

Fill the air gap with the liquid. Ring diameters shrink because the effective wavelength inside the medium is λ/μ . Comparing diameters with and without liquid:

$$\mu = D_{n(\text{air})}^2 / D_{n(\text{liquid})}^2$$

TRICK: "Centre is always DARK" — remember this instantly using the extra π phase-shift on reflection from the denser glass plate below.

✓ QUICK REVISION

- Circular fringes from interference in a variable-thickness air film between lens & plate.
- Centre is dark (extra $\lambda/2$ path due to phase reversal at glass plate).
- $D_n \propto \sqrt{n}$ (dark rings), so ring spacing decreases as n increases.

- Used to find λ of light and μ of liquids experimentally.

Rayleigh Criterion (Resolving Power)

When light from a point source passes through a circular aperture (like a telescope or microscope lens), diffraction spreads it into a small bright disc surrounded by faint rings, called the *Airy pattern*, instead of a perfect point. This limits how well an optical instrument can distinguish (resolve) two closely-spaced objects.

► Statement of Rayleigh's Criterion

"Two nearby point sources are said to be just resolved when the central maximum of the diffraction pattern of one source falls exactly on the first minimum of the diffraction pattern of the other source."

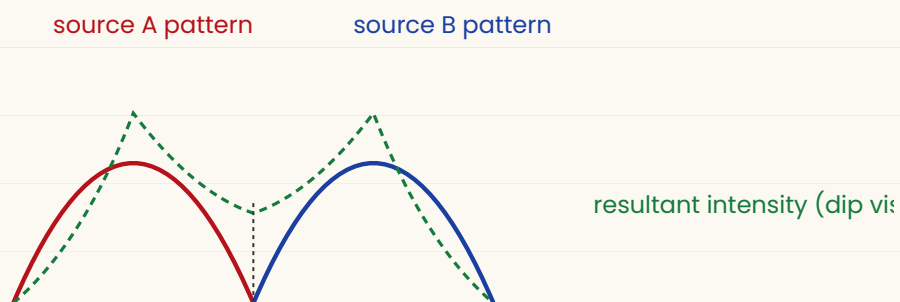


fig — Rayleigh criterion: central max of A coincides with first min of B

► Minimum Resolvable Angle (for a circular aperture of diameter a)

The first minimum of the diffraction pattern of a circular aperture occurs at an angle:

$$\theta_{\min} = 1.22 \lambda / a$$

where λ = wavelength of light, a = diameter of the aperture (lens/objective). This $\theta_{\min}$ is the smallest angular separation two points can have and still be resolved.

► Resolving Power

Resolving power is defined as the reciprocal of the minimum resolvable angle:

$$\text{Resolving Power} = 1/\theta_{\min} = a / 1.22\lambda$$

Condition	Result
Separation between patterns $> \theta_{\min}$	Well resolved (two clear points)
Separation $= \theta_{\min}$	Just resolved (a small dip visible)
Separation $< \theta_{\min}$	Not resolved (appears as one blur)

★ **EXAM TIP:** Larger aperture (a) or shorter wavelength (λ) $\Rightarrow$ smaller $\theta_{\min} \Rightarrow$ better resolving power. This is why telescopes use large objective lenses/mirrors.

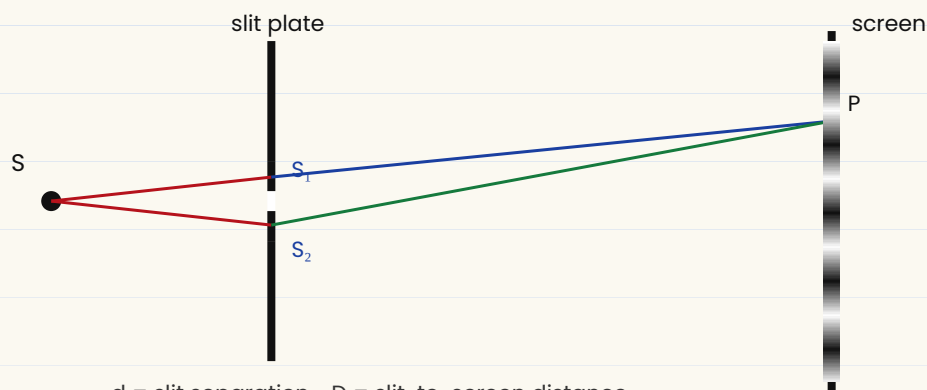
TRICK: *"Big Lens, Sharp Sense" — a bigger aperture gives sharper (better) resolution.*

✓ **QUICK REVISION**

- *Rayleigh criterion: central max of one pattern on first min of the other = "just resolved".*
- *$\theta_{\min} = 1.22\lambda/a$ for a circular aperture.*
- *Resolving Power = $1/\theta_{\min}$ — increases with larger aperture, decreases with longer λ .*

Young's Double Slit Experiment (YDSE) & Fringe Width

YDSE is the classic experiment that proved light behaves as a wave — it produces alternate bright and dark bands (fringes) on a screen due to interference of light from two narrow, coherent slits.



d = slit separation, D = slit-to-screen distance

fig — YDSE setup: source S , slits S_1, S_2 , screen with fringes

► Path Difference at Point P on Screen

Let d = separation between slits, D = distance of screen from slits, x = distance of P from centre O . For $D \gg d$, the path difference is:

$$\Delta = S_2P - S_1P = x d / D$$

► Conditions for Bright and Dark Fringes

Condition	Result	Position x_n
$\Delta = n\lambda$	Bright fringe	$x_n = n\lambda D / d$
$\Delta = (2n-1)\lambda / 2$	Dark fringe	$x_n = (2n-1)\lambda D / 2d$

► Fringe Width (β)

Distance between two consecutive bright (or two consecutive dark) fringes.

$$\beta = \lambda D / d$$

Note: fringe width is the same for bright and dark fringes — so fringes are equally spaced, unlike Newton's Rings!

★ **EXAM TIP:** $\beta \propto \lambda$ and $\beta \propto D$, but $\beta \propto 1/d$. If d is doubled, fringe width becomes half — a very common numerical.

TRICK: "D on top, d on bottom" — $\beta = \lambda D / d$ — remember which one is numerator vs denominator by this phrase.

✓ **QUICK REVISION**

- *Path difference at $\mathcal{P}$: $\Delta = x d / \mathcal{D}$*
- *Bright: $\Delta = n\lambda$; Dark: $\Delta = (2n-1)\lambda/2$*
- *Fringe width $\beta = \lambda \mathcal{D} / d$ — equally spaced fringes.*

Mach–Zehnder Interferometer

The Mach–Zehnder Interferometer (MZI) splits a beam of light into two separate paths using beam-splitters, sends them along two different arms, and then recombines them to produce interference. It is widely used to measure small changes in refractive index / phase (e.g. in gas flow, plasma, and heat-transfer studies).

► Construction & Working

It uses two beam-splitters (BS_1 , BS_2) and two fully-silvered mirrors (M_1 , M_2) arranged in a rectangle/square:

- Light from the source hits BS_1 , which splits it into two beams of equal intensity — one transmitted (Path 1) and one reflected (Path 2).
- Path 1 travels to mirror M_1 , then to the second beam-splitter BS_2 .
- Path 2 travels to mirror M_2 , then also to BS_2 .
- BS_2 recombines both beams, and interference fringes are observed at the two output detectors (D_1 and D_2).

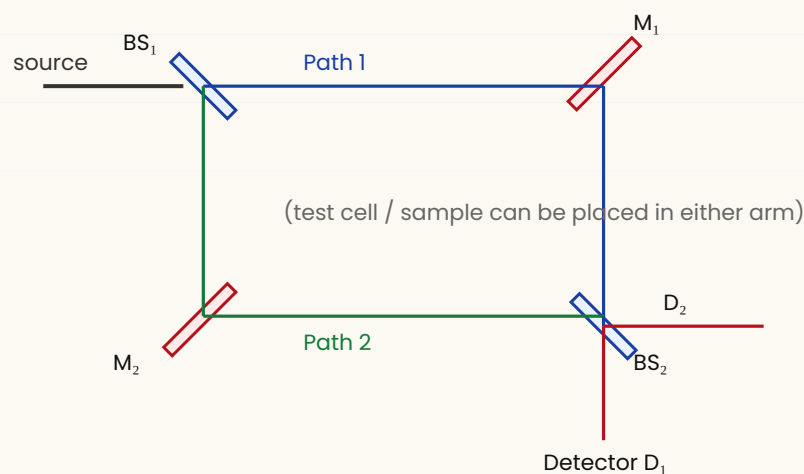


fig — Mach–Zehnder Interferometer schematic

► Working Principle

If both arms (paths) are perfectly identical, the recombined beams interfere constructively and simply reproduce the input beam. If a sample (gas, transparent object, or phase object) is placed in one arm, it changes the optical path length of that arm slightly. This produces a measurable phase difference, which shifts the interference fringes — allowing precise measurement of the sample's refractive index or thickness.

★ **EXAM TIP: Unlike Michelson's interferometer (Topic 5) which folds the beam back, MZI separates the two beams completely so a test object can be placed in only one path — a key comparison point often asked.**

► Applications

- Measuring refractive index of gases and transparent samples.
- Flow visualization (schlieren-type studies), heat transfer & plasma density studies.

- *Used in quantum optics and fibre-optic sensing.*

TRICK: "Mach = two Mirrors, two Beam-splitters, in a Maze" — MZI has 2 BS + 2 M forming a rectangular maze-like path.

✓ **QUICK REVISION**

- *Uses 2 beam-splitters + 2 mirrors to split light into two separate arms and recombine it.*
- *A sample placed in one arm creates a phase shift → fringe shift.*
- *Used to measure refractive index, flow, and phase objects precisely.*

Michelson Interferometer

The Michelson Interferometer splits a beam into two, sends them along two perpendicular arms which reflect off mirrors and come back to recombine at the same beam-splitter — producing interference. It was famously used in the Michelson–Morley experiment to test for the "ether".

► Construction

- S — monochromatic light source.
- G_1 — a half-silvered (partially reflecting) glass plate set at 45° , called the beam-splitter. It splits incoming light into a reflected beam and a transmitted beam.
- M_1 — a fixed, fully-silvered plane mirror (one arm).
- M_2 — a movable, fully-silvered plane mirror mounted on a micrometer screw (other arm).
- G_2 — a "compensating plate" (identical to G_1 but unsilvered), added so both beams pass through the same thickness of glass — making the two optical paths truly comparable.

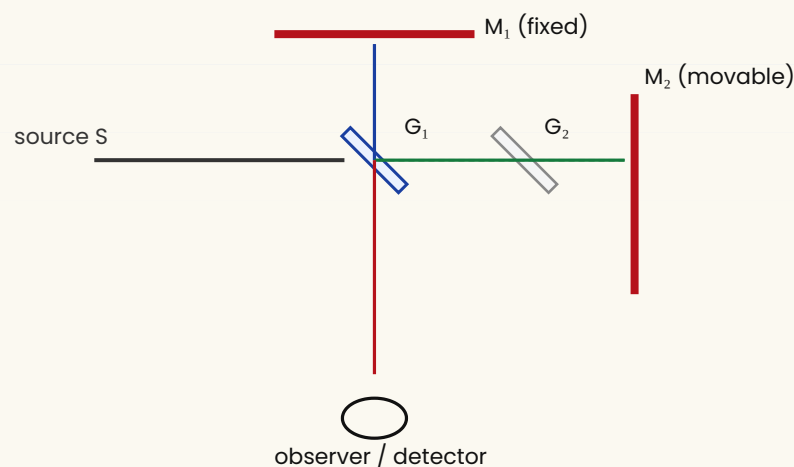


fig — Michelson Interferometer schematic

► Working

- Light from S hits G_1 and splits into two beams of equal intensity (one reflected up towards M_1 , one transmitted towards M_2).
- Both beams reflect back from their mirrors, return to G_1 , and recombine — travelling together towards the observer, where interference fringes are seen.
- Moving mirror M_2 slightly changes the path length of that arm, which shifts the fringe pattern.

► Path Difference & Fringe Shift

If M_2 is moved a distance d , the path length of that arm changes by $2d$ (light travels the extra distance twice — going and coming back). The path difference changes by:

$$\Delta = 2d$$

Each time the mirror moves by $\lambda/2$, exactly one fringe shifts across the field of view. So by counting the number of fringes ($\mathcal{N}$) that shift for a mirror movement d , we can find the wavelength:

$$\lambda = 2d / \mathcal{N}$$

★ **EXAM TIP:** " $\lambda = 2d/\mathcal{N}$ " is the standard numerical formula — used to calculate wavelength from the number of fringes crossing while moving the mirror by a measured distance d . Practice this numerical!

► Applications

- Precise measurement of wavelength of light.
- Measuring very small distances / thickness (using known λ).
- Michelson–Morley experiment (famous for disproving the "ether" hypothesis).
- Basis of LIGO — used to detect gravitational waves.

TRICK: "One Mirror Moves, One Stays" — $\mathcal{M}_2$ moves (measurement arm), $\mathcal{M}_1$ stays fixed (reference arm); G_2 only "compensates" — it never reflects the main beams.

✓ QUICK REVISION

- G_1 = beam splitter, G_2 = compensating plate, $\mathcal{M}_1$ = fixed mirror, $\mathcal{M}_2$ = movable mirror.
- Moving $\mathcal{M}_2$ by d changes path by $2d$.
- $\lambda = 2d/\mathcal{N}$ — used to measure wavelength precisely.
- Famous use: Michelson–Morley experiment; modern use: LIGO gravitational wave detection.

Unit 2 — Formula Sheet (Last-Minute Revision)

Topic	Key Formula
Newton's Ring — dark rings	$D_n = \sqrt{4n\lambda R}$
Newton's Ring — bright rings	$D_n = \sqrt{2(2n-1)\lambda R}$
Newton's Ring — find λ	$\lambda = (D_q^2 - D_p^2)/4R(q-p)$
Rayleigh Criterion	$\theta_{\min} = 1.22\lambda/a$; $R.P. = 1/\theta_{\min}$
YDSE — path difference	$\Delta = x d/D$
YDSE — fringe width	$\beta = \lambda D/d$
Michelson — wavelength	$\lambda = 2d/N$

► Frequently Asked (Previous-Year Style) Questions

- Describe the experimental arrangement of Newton's Rings and derive the expression for the diameter of dark rings.
- Why is the central spot in Newton's Rings dark? Explain with reasoning.
- How would you determine the wavelength of light / refractive index of a liquid using Newton's Rings?
- State and explain Rayleigh's Criterion for resolving power. Derive $\theta_{\min} = 1.22\lambda/a$.
- Describe Young's Double Slit Experiment and derive the expression for fringe width.
- Explain the construction and working of a Mach-Zehnder Interferometer with a labelled diagram.
- Explain the construction and working of a Michelson Interferometer. How is it used to determine wavelength of light?

★ **OVERALL EXAM TIP: Numericals on Newton's Rings (finding λ or μ), YDSE fringe width, and Michelson's $\lambda = 2d/N$ are asked almost every year — practice at least 3–4 numericals from each.**

MASTER MNEMONIC — "N-R-Y-M-M": Newton's ring → Rayleigh → YDSE → Mach-Zehnder → Michelson — the natural order of Unit 2, as usually taught & asked.

✓ UNIT 2 IN ONE LINE

Wave Optics shows how light waves interfere to form patterns — circular fringes in Newton's Rings, diffraction limits in Rayleigh's criterion, straight equally-spaced fringes in YDSE, and precise phase/path measurement using the Mach-Zehnder and Michelson interferometers.